MISO OmniSearch

Master Technical Showcase Script & Engineering Runbook

Target Audience:	MISO Engineering, Cloud Infra (Azure), Data & Architecture Teams	**Target Run Time:**	15–20 Minutes (Interactive Q&A; Focus)
Format:	Screen Cues, Talking Tracks, Tech Deep Dive & Live Probes	**Verification:**	**47 / 47 Unit Tests Passing (100% OK)**

■ PRE-FLIGHT TECHNICAL CHECKLIST (Run 5 Min Before Call)

Component	Launch Command / Script	Target Endpoint / Verification	Status
FastAPI Backend	python -m uvicorn backend.main:app --port 8000	http://localhost:8000/healthz → 200 OK	**READY (Port 8000)**
React Frontend	cd frontend && npm run dev	http://localhost:5173 → Pulsing SSE badge	**READY (Port 5173)**
Automated Tests	python -m unittest discover -s backend -v	47 tests run & verified in < 25 seconds	**47/47 PASSED**
Browser & DevTools	Chrome / Edge @ 100% zoom with Network Tab open	Inspect X-Correlation-ID & X-Response-Time-Ms	**ACTIVE**

SECTION 1: Welcome & The Problem We Tackled

0:00 – 3:00 (3 min)

■ Screen Cue: Landing Page Display

Share screen showing OmniSearch landing page (http://localhost:5173). Do not type yet. Let the MISO team observe the clean interface, Session Radar telemetry banner, and MISO Navy design tokens.

■■ PRESENTER TALKING TRACK:

"Good morning, everyone. Thank you so much for having us back. We're excited to speak with MISO's developers, system architects, and technical team today.

When people hear about AI in 2026, the immediate assumption is usually a generic chatbot wrapper. But anyone who works with wholesale power markets knows that **generic LLMs are dangerously unsuited for grid operations**. If a chatbot hallucinates a decimal point on a Locational Marginal Price, gets confused by Daylight Saving Time, or cites an outdated tariff attachment, it creates real regulatory liability and financial risk.

When we approached **MISO Challenge Prompt 1 — Intelligent Navigation of MISO's Public Information**, we analyzed the friction that external market participants face every day on misoenergy.org:
• **The Legacy CSV Retirement Chasm:** MISO is modernizing its data stack by deprecating legacy flat CSV reports like rt_lmp_final.csv and moving to the Azure API Management Data Exchange portal. But hundreds of municipal co-ops, trading desks, and utility billing systems rely on those CSV schemas and have struggled to migrate without clear data crosswalks.
• **The 75+ Acronym Fortress:** Wholesale markets are dense. Terms like LMP, CONE, PRA, MTEP, LRTP, JTIQ, and ICCP create immense barriers for journalists, state regulators, and municipal utilities.
• **The Support Queue Bottleneck:** Because public grid data is buried in 1,000-page PDF Business Practice Manuals and fragmented web pages, over **12,500 routine data tickets** flood MISO customer support every year. That takes 24 to 72 hours per ticket and costs MISO roughly **$3.75 Million** in diverted engineering and analyst time annually.

Our team engineered **MISO OmniSearch**: an enterprise-grade predictive context and comparative knowledge engine built on four non-negotiable principles: **Zero Cold Start**, **Zero Hallucination**, **Role-Based Analytical Depth**, and **Enterprise Production Readiness**."

SECTION 2: Live Prototype Walkthrough

3:00 – 8:00 (5 min)

■ Scene 1: Zero Cold Start & Live SSE Telemetry

Point cursor to top SessionRadar banner. Highlight pulsing green badge: '● Live SSE Stream (88,450 MW)'.

■■ PRESENTER TALKING TRACK:

"Notice what's happening on screen right now. I haven't typed a single keystroke, yet the page is already alive with verified telemetry. Most search systems suffer from 'blank search bar syndrome.' In OmniSearch, our **Session Radar** senses the user's workflow context. If they arrived from market reports, it has already pre-loaded today's average LMP spreads, cleared volumes, and quick-start research chips.

Notice this glowing green badge: ● Live SSE Stream. Rather than having thousands of browser clients hammering MISO's public operations endpoint with aggressive HTTP polling, our backend maintains an asynchronous **Server-Sent Events (SSE)** stream. When MISO's public API pushes a new 5-minute fuel interval, our backend caches it with a 60-second TTL and fans it out over a persistent connection with zero frontend re-fetching overhead."

■ Scene 2: Predictive Search & 360° Knowledge Canvas

Press Ctrl + K (spotlight activates). Type 'Indiana Hub'. Observe sub-50ms autocomplete categorization by Hub and Metric. Press Enter. Knowledge Canvas renders.

■■ PRESENTER TALKING TRACK:

"Now let's execute a search. I'll hit Ctrl + K and type 'Indiana Hub'. Notice the sub-50ms autocomplete categorization. When I hit Enter, OmniSearch compiles an interactive **360° Knowledge Canvas**:
• At the top, an **authoritative direct answer** explaining price formation and peak-hour volatility.
• An immutable **source citation badge** pointing to MISO's July 2025 Corporate Fact Sheet and Tariff Module C.
• **Four isolated KPI metric cards**: Real-Time average ($38.45/MWh), Day-Ahead average ($37.20/MWh), Peak Hour interval, and 24-hour cleared volume.
• An interactive **Recharts curve** plotting 24 hours of hourly LMP pricing with toggleable **3-Part LMP Decomposition**: LMP = MEC (Energy) + MCC (Congestion) + MLC (Loss).
Every number you see here is mathematically grounded. If an engineer prefers tabular inspection, one click toggles into an accessible data table formatted with IBM Plex Mono tabular numerals."

■ Scene 3: Dynamic Persona Inference & Grounded Copilot Drawer

Switch persona dropdown to 'Public / Media'. Type search query: 'HE 19 arbitrage spread volatility'. Observe recommendation banner slide in. Click 'Switch to Power Trader'. Then click 'Chat with this Canvas'.

■■ PRESENTER TALKING TRACK:

"One of our most unique features is our **Dynamic Persona Inference Engine**. A power trader analyzing arbitrage spreads needs completely different intelligence than a municipal co-op board member or a journalist.

Watch what happens: I am currently in 'Public / Media' mode. I'll search for a trading term: 'HE 19 arbitrage spread volatility'. OmniSearch parses the lexical syntax, recognizes the domain pattern, and recommends switching to Power Trader. With one click, the canvas instantly re-shapes itself to focus on congestion costs and peak ramp hours with zero page reload.

And clicking **'Chat with this Canvas'** opens our slide-out **Canvas Copilot Drawer** powered by Gemini 3.6 Flash. The model is injected with active canvas state, but its outputs are verified by our **numerical grounding firewall**—rejecting any hallucinated numbers outside official market bounds."

■ Scene 4 & 5: Comparison Engine & 1-Page PDF Briefing Studio

Open Comparison workspace. Select Indiana.Hub vs. Michigan.Hub vs. Texas.Hub. Observe synchronized curves. Then click '1-Page PDF Fact Sheet' (downloads in < 2.5s). Open PDF. Press Ctrl + J to show Jargon HUD.

■■ PRESENTER TALKING TRACK:

"With our **Comparison Engine**, stakeholders can benchmark up to 6 regional commercial hubs side-by-side with synchronized 24-hour dual-axis curves. You immediately see Michigan Hub trading at a **+$4.40/MWh premium** over Indiana Hub due to northern transmission constraints.

When executives or board members need physical handouts, clicking **'Download Briefing PDF'** compiles a branded, publication-ready executive briefing in under 2.5 seconds using pure Python **ReportLab**. The document is mathematically budgeted to guarantee **strictly 1 physical page** with zero spillover.

And pressing Ctrl + J activates our **Universal Jargon HUD** containing **75+ verified energy acronyms**, featuring both plain-English ELI5 explanations and exact technical tariff formulas."

SECTION 3: The 5 Enterprise Additions Built for MISO's Tech Stack — 8:00 – 13:00 (5 min)

■ Feature 1: Data Contract & Market Anomaly Firewall

Click on the '■■ Data Contract: 100% Verified' badge in SessionRadar. Inspect DataContractModal showing 4 validation rules and SHA-256 ingestion hash.

■■ PRESENTER TALKING TRACK:

"First: **The Data Contract & Market Anomaly Firewall** in backend/data_validator.py. Wholesale electricity data cannot tolerate corrupted payloads. We implemented automated Pydantic data contracts:
1. **LMP Conservation Contract:** Verifies physical equality LMP = MEC + MCC + MLC with a tight ±$0.05 rounding tolerance across all 144 daily hub intervals.
2. **Generation Mass Balance:** Enforces physical non-negativity across all fuels and validates total generation within MISO's 20 GW – 160 GW envelope.
3. **Telemetry Continuity:** Detects stale feeds if interval gaps exceed 5 minutes.
4. **Market Anomaly Scanner:** Detects negative LMPs from localized renewable curtailment and regional congestion spikes.
Every audit payload is cryptographically sealed with a **SHA-256 ingestion hash** for regulatory audit trails."

■ Feature 2: Hybrid Semantic RAG for BPM Rulebooks

Click '■ BPMs (RAG)' button in header. Type 'LMP formula'. See BPM 002 §4.3.1 card. Click suggestion chip 'DPP cluster cycle' to show BPM 015 §3.1.

■■ PRESENTER TALKING TRACK:

"Second: **Hybrid Semantic RAG for Business Practice Manuals (BPMs)** in backend/bpm_service.py. MISO governs wholesale markets through over 20 detailed BPMs. Traditional vector search often retrieves generic text rather than exact clauses.

We built a **Hybrid Retrieval Engine** combining **BM25 lexical token matching** with **semantic concept scoring**. Searching 'LMP formula' directly extracts **BPM 002 §4.3.1** with the exact decomposition equation and direct ZIP download link. Searching 'DPP cluster cycle' yields **BPM 015 §3.1** on interconnection milestone payments (M2/M3)."

■ **Feature 3: Microsoft Entra ID (Azure AD) SSO & RBAC Middleware**

Click the top-right SSO role simulator dropdown ('■■ SSO: PowerTrader ▼'). Switch to 'MISO.StateRegulator'. UI persona and permissions synchronize instantly.

■■ PRESENTER TALKING TRACK:

"Third: **Microsoft Entra ID (Azure AD) SSO and Role-Based Access Control** in backend/auth_middleware.py. MISO's production infrastructure lives in Microsoft Azure. Our authentication layer decodes OIDC Bearer tokens and extracts the roles claim.

In our live simulator, switching from MISO.PowerTrader to MISO.StateRegulator automatically updates claims, unlocks compliance logs and transmission cost allocation, and adapts the UI without a page reload. In production, this enforces fine-grained rate limits across staff, market participants, and the public."

■ **Feature 4: Cloud-Native Observability & Kubernetes Probes**

Click '■■ DevOps / SRE' button in header. Show Tab 1 (Liveness & Readiness Probes) and Tab 2 (Prometheus Metrics). Open browser DevTools Network tab to reveal headers.

■■ PRESENTER TALKING TRACK:

"Fourth and fifth: **Cloud-Native Observability and Live SSE Streaming** in backend/observability.py. To run smoothly on Azure Kubernetes Service (AKS), we expose:
• GET /healthz (Liveness Probe): Validates the FastAPI ASGI process is alive.
• GET /readyz (Readiness Probe): Validates downstream MISO operations feeds and data stores before traffic is routed.
• GET /metrics: Standard Prometheus 0.0.4 text exposition tracking latencies, cache hit ratios, active SSE subscribers, and grounding verification rates.
Our middleware also injects X-Correlation-ID and X-Response-Time-Ms headers on every request for distributed tracing across Azure Application Insights."

SECTION 4: Technical Deep Dive & System Architecture — 13:00 – 16:00 (3 min)

■ **Visual Screen Cue: Architecture Diagram**

Switch to secondary browser tab showing the C4 Container Architecture Diagram (from docs/miso-tech-team-handbook.md).

■■ PRESENTER TALKING TRACK:

"Let's look under the hood at our architecture and data flow:

1. Technology Stack: FastAPI + Pydantic v2 on Python 3.11 for async concurrency and sub-millisecond serialization; React 18 + Vite + TypeScript in strict mode for type safety; and ReportLab for headless, ultra-fast PDF compilation (< 2.5s, < 15MB RAM).

2. Ingestion & Caching Layer: In miso_client.py, we ingest 5-minute fuel telemetry from public-api.misoenergy.org/api/FuelMix. To respect MISO's rate-limit policy (≤ 1 req/min), we implement an in-memory 60-second TTL cache with graceful fallback to our baseline corporate dataset if MISO's endpoint times out.

3. Dual-Engine AI & NERC CIP Compliance: In llm_service.py, we support cloud-based Google Gemini 3.6 Flash alongside local on-premise **Ollama** (running llama3.2:3b or mistral). For control room environments governed by **NERC CIP**, this guarantees that zero electric system data or operator queries ever traverse the public internet."

SECTION 5: Limitations, Assumptions & Honest Trade-offs

16:00 – 18:00 (2 min)

■ **Visual Screen Cue: Presenter Face-to-Face**

Stop sharing or keep architecture slide visible. Speak candidly and collegially with the engineering team.

■■ **PRESENTER TALKING TRACK:**

"As engineers, we want to be completely transparent about our current prototype's assumptions and limitations:

1. **In-Memory Cache vs. Distributed Redis:** The 60-second TTL cache currently lives in Python process memory. In a multi-replica AKS cluster, this would cause cache fragmentation. In production, this should be backed by **Azure Cache for Redis**.
2. **Commercial Hub Pricing Baseline:** Without private Azure APIM production keys during development, our 24-hour hub pricing models are derived from verified baseline curves rather than a live 5-second ICCP feed. However, miso_client.py already includes the exact Ocp-Apim-Subscription-Key header scaffolding ready to connect.
3. **Static BPM Indexing vs. Continuous Ingestion:** Our BPM hybrid search indexes 20 key manuals. In production, we envision an automated pipeline ingesting new FERC filings and BPM PDF revisions from MISO's CDN into **Azure AI Search** on a weekly schedule."

SECTION 6: Collaborative Production Scaling Roadmap on Azure

18:00 – 20:00 (2 min)

■ **Visual Screen Cue: Production Scaling Blueprint**

Show Azure Production Architecture Diagram from docs/miso-tech-team-handbook.md.

Phase	Target Azure Technology	Production Architecture Role
Phase 1: Compute	Azure Kubernetes Service (AKS) + Azure Front Door	Deploy FastAPI container with HPA (3 to 20 pods); React on Static Web Apps with global edge CDN.
Phase 2: Data & State	Azure Cache for Redis + TimescaleDB (PostgreSQL)	Sub-millisecond distributed caching across pods; partitioned time-series storage for 365-day historical curves.
Phase 3: Telemetry	Azure Event Hubs / Kafka Ingestion Workers	Stream live 5-second SCADA, ICCP, and State Estimator feeds directly from MISO EMS into cache layer.
Phase 4: Security	Microsoft Entra ID + Azure Key Vault + mTLS	Enterprise OIDC RBAC claims enforcement, HSM key storage, and end-to-end NERC CIP compliant networking.

■■ **PRESENTER TALKING TRACK:**

"**Closing Summary:** By deflecting over **80% of routine data inquiries**, MISO OmniSearch saves MISO over **$3.0 Million annually** and returns **40,000 engineering hours** back to grid reliability and transmission planning.

All 47 automated tests are passing, our build is clean, and the codebase is pushed to GitHub. We would love to open the floor, hear your thoughts, and discuss how we can build upon this together. Thank you!"

SECTION 7: Anticipated Tech Team Q&A; Cheat Sheet

5+ Minutes

■ **Tech Question 1: "How do you guarantee the LLM won't hallucinate a price during a high-volatility event?"**

Key Defense: We use a two-tier defense. First, prompt context is strictly bounded to verified JSON numbers. Second, llm_service.py executes _verify_numerical_grounding via post-generation regex. It scans every price and megawatt figure against the ground-truth dictionary. If any number deviates outside official market bounds, the output is rejected and falls back to our deterministic template.

■ **Tech Question 2: "Why did you implement Server-Sent Events (SSE) instead of WebSockets?"**

Key Defense: WebSockets are bidirectional and require stateful connection tracking and custom protocol framing. For public grid telemetry, data flow is purely unidirectional from MISO down to the client every 5 minutes. SSE operates over standard HTTP/2, works natively through corporate proxies and firewalls, supports automatic reconnection out of the box, and has significantly lower server memory overhead.

■ Tech Question 3: "How does your legacy crosswalk help external developers who are upset about retiring CSV files?"

Key Defense: In crosswalk.json, we mapped legacy CSV column headers like HE, NODE_NAME, and LMP_CONG to modern REST endpoints on data-exchange.misoenergy.org. When a developer searches for an old CSV name, OmniSearch displays the exact JSON path, Swagger endpoint, and timezone conversion notes (EPT vs UTC) so they can rewrite their scripts in minutes.

■ Tech Question 4: "What is the memory and CPU footprint of your ReportLab PDF generator?"

Key Defense: Unlike headless Chrome or Puppeteer, which require launching a 200MB Chromium process that introduces memory spikes and CVE attack vectors, ReportLab compiles pure Python byte streams in memory. It uses less than 15MB of RAM per worker and compiles the full 1-page PDF in under 2.5 seconds.